DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE



Statistics - 1

Lecture No. 11



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Recap of previous lecture



Topic NORMAL DISTRIBUTION

Topics to be Covered



Topic

(1) CURVE FITTING

(2) Correlation & Regression.

Curve fitting

① Line of Best fit



Consider a set of points given as follows;

x_1	x_2	x_3	x_4	x_5	- - - - -	x_n
y_1	y_2	y_3	y_4	y_5	- - - -	y_n

We have assumed that line of best fit is $\boxed{y=a+bx}$ ——— (1)

Here Constant a & b can be calculated by solving following equ^{n}s;

$$\Sigma y = na + b\Sigma x \quad \text{---- (2)}$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \quad \text{---- (3)}$$

} these two equ^{n}s are called Normal equ^{n}s to find line of Best fit.

$y = f(x)$ ——— Target.

Let $E_i = y_i - f(x_i)$

$E_i^2 = (y_i - f(x_i))^2$

$u = (y_i - f(x_i))^2$

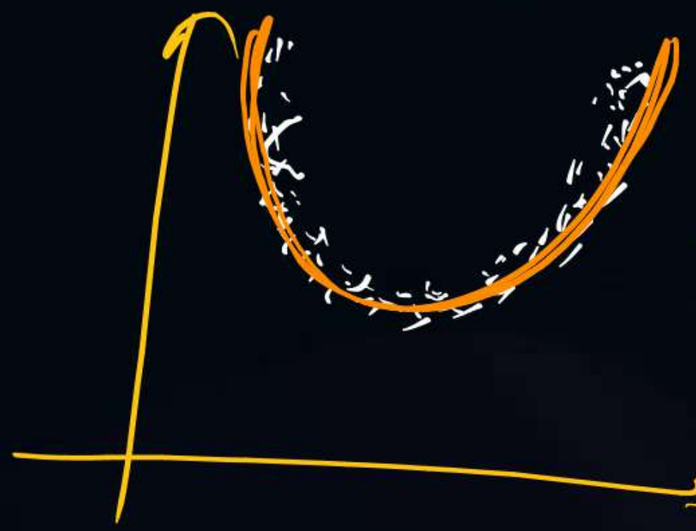
$\frac{\partial u}{\partial x} = 0$ & $\frac{\partial u}{\partial y} = 0$

$\Rightarrow \Sigma y = na + b\Sigma x$

& $\Sigma xy = a\Sigma x + b\Sigma x^2$

this process is called Method of least squares

(2) Parabola of Best fit →



x_1	x_2	x_3	- - - -	x_n
y_1	y_2	y_3	- - - -	y_n

Let the parabola of Best fit is

$$y = a + bx + cx^2$$

↳ to find a, b, c, we will solve Normal eqn's ;

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$
$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

} 3 eqn's

Q: Find the line of Best fit for the following Data ?

x	1	2	3	4	5
y	3	5	7	9	11

Method I Using observation it is $y = 2x + 1$ this is the line of Best fit.

Method II

x	y	x^2	xy
1	3	1	3
2	5	4	10
3	7	9	21
4	9	16	36
5	11	25	55
Σ 15	35	55	125

No. of points in our table $n = 5$

Let the line of Best fit is $y = a + bx$ — (1)

where a & b can be Calculated by solving following Normal eqn's.

$\Sigma y = na + b\Sigma x$ & $\Sigma xy = a\Sigma x + b\Sigma x^2$

$35 = 5a + b(15)$ & $125 = a(15) + b(55)$

$\Rightarrow a = 1, b = 2$ Hence by (1), line of Best fit is $y = 1 + 2x$

Q: Find the line of Best fit for the following Data :

1	2	3	4	5	6	7
9	8	10	12	11	13	14

(M-I) Using observation :- Not easy to crack.

(M-II)

x	y	x^2	xy
1	9		
2	8		
3	10		
4	12		
5	11		
6	13		
7	14		
$\Sigma =$			

HW:

- - - - - - - - - -
- - - - - - - - - -

Ans: $y = 7.28 + 0.928x$

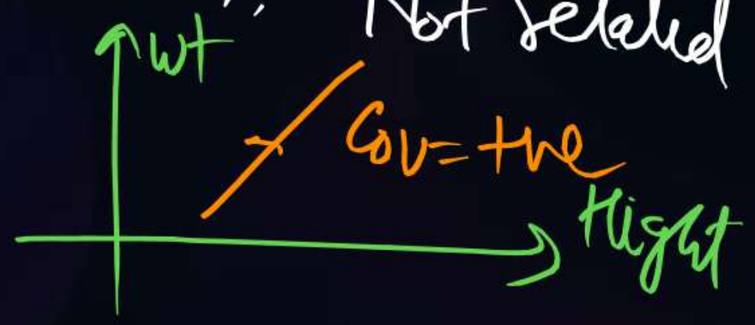
Covariance – It measures the simultaneous variation of two Random Variables & it is defined as $\boxed{Cov(X,Y) = E(XY) - E(X)E(Y)}$, $-\infty < Cov(X,Y) < \infty$

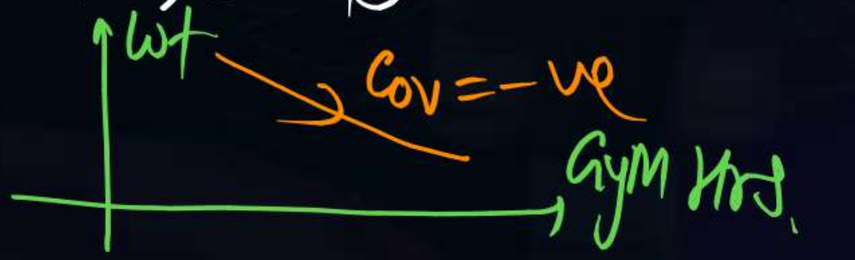
Correlation – It measures the simultaneous variation of two Random Variables & Correlation Coeff^{n} (r) is defined as $\boxed{r = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}}$ $-1 \le r \le 1$

If two variables are perfectly correlated in the +ve sense then $r = 1$

" " " " " " in -ve " then $r = -1$

" " " " Not related then $r = 0$





Covariance

(1) It measures the direction of relationship

(2) In different unit, Cov is diff

(6) It doesn't tell, how strong the relationship is ?

(3) if X & Y are Ind then $Cov(X,Y)=0$

(4) $-\infty < Cov(X,Y) < \infty$

(5) $Cov(X,Y) = E(XY) - E(X)\cdot E(Y)$

Correlation

(1) It measures the direction as well as the strength of Relationship.

(2) In diff units, Correlation Coeff is not diff.

(6) It tells the strength of relationship

(3) If X & Y are Ind then $\rho = 0$

(4) $-1 \le \rho \le 1$

(5) $\rho = \dfrac{Cov(X,Y)}{\sigma_X \sigma_Y}$

REGRESSION :-

If x & y are two Random Variables then we do not have any idea which one is Cause and which one is effect then we will develop two Regression lines as follows;

R. line y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where $b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$ = R Coeff of y on x

R. line x on y :-

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where $b_{xy} = r \frac{\sigma_x}{\sigma_y}$ = R Coeff of x on y

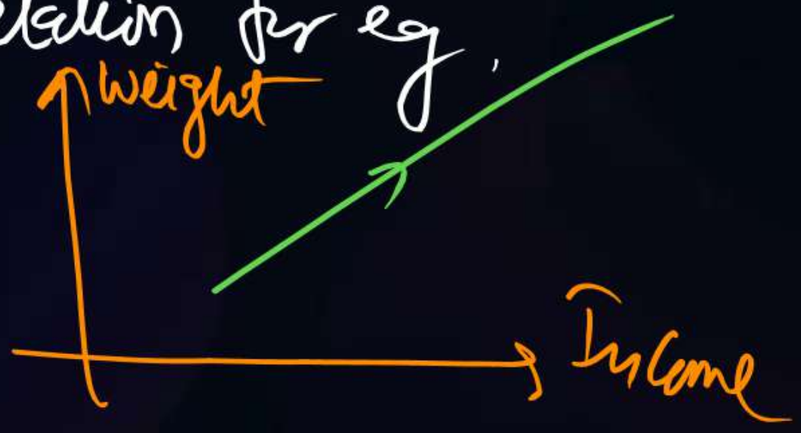
i.e Regression Concept tells the variation of one variable with Respect to another variable.

CORRELATION

(1) It measures the Direction as well as Strength of Relationship,

(2) It has nothing to do with Cause & effect

(3) Sometimes there may be senseless Correlation for eg.



REGRESSION

(1) It Measures the Cause & effect Relationship b/w two Random Variables

for eg if Supply is Constant then

Case I: Price $\propto$ Demand

Case II: Demand $\propto \frac{1}{\text{Price}}$

(2) It measures the changes in Dep Variable w.r. to Independent Variable

(3) There is No senseless Relation b/w x & y

Properties:- (1) Intersecting Point of two Regression Lines is nothing but $(\bar{x}, \bar{y})$

(2) b_{yx}, b_{xy} and r have same sign (i.e either all +ve or all -ve).
($\because$ r is the GM of b_{yx} & b_{xy} & we know that all 3 have same sign)

(3) r is the G.M of b_{yx} & b_{xy} i.e $\boxed{r = \pm\sqrt{b_{yx}\, b_{xy}}}$

Proof: $b_{yx}\, b_{xy} = \left(r\frac{\sigma_y}{\sigma_x}\right)\left(r\frac{\sigma_x}{\sigma_y}\right) = r^2 \Rightarrow r = \pm\sqrt{b_{yx}\, b_{xy}}$

(4) if one R. Coeff is > 1 then other R. Coeff must be less than 1 ($\because -1 \le r \le 1$)
i.e both can be < 1 also But Both can't be > 1 simultaneously

Sp. Note:- GM of two +ve Nos a & $b = +\sqrt{ab}$

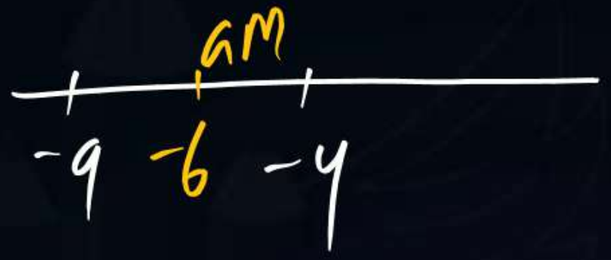
GM of two -ve Nos a & $b = -\sqrt{ab}$

GM of one +ve & one -ve No = N.D.

e.g. GM of 4 & $9 = ? = \sqrt{4\times 9} = \sqrt{36} = +6$



e.g. GM of -9 & $-4 = ? = -\sqrt{(-9)(-4)} = -\sqrt{36} = -6$



e.g. GM of -9 & $4 = ?$ = Not Defined

(5) Angle b/w two R. lines is $\tan\theta = \left(\frac{1-r^2}{r^2}\right)\cdot\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$

Q. Find Correlation Coeff, Regression Coeff, R. Line y on x & R. Line x on y for

1	2	3	4	5
3	5	7	9	11

$n=5$

Sol:

	x	y	x^2	xy	y^2
	1	3	1	3	9
	2	5	4	10	25
	3	7	9	21	49
	4	9	16	36	81
	5	11	25	55	121
$\Sigma =$	15	35	55	125	285

$$\bar{x} = E(x) = \frac{\Sigma x}{N} = \frac{15}{5} = 3$$

$$\bar{y} = E(y) = \frac{\Sigma y}{N} = \frac{35}{5} = 7$$

$$\overline{xy} = E(xy) = \frac{\Sigma xy}{N} = \frac{125}{5} = 25$$

$$\bar{x^2} = E(x^2) = \frac{\Sigma x^2}{N} = \frac{55}{5} = 11$$

$$\bar{y^2} = E(y^2) = \frac{\Sigma y^2}{N} = \frac{285}{5} = 57$$

$$Cov(xy) = E(xy) - E(x)\cdot E(y) = 25 - (3)(7) = 4$$

$$Var(x) = E(x^2) - E^2(x) = 11 - (3)^2 = 2, \quad \sigma_x = \sqrt{2}$$

$$Var(y) = E(y^2) - E^2(y) = 57 - (7)^2 = 8, \quad \sigma_y = 2\sqrt{2}$$

① Correlation Coeff $r = \frac{Cov(x,y)}{\sigma_x \sigma_y} = \frac{4}{(\sqrt{2})(2\sqrt{2})} = 1$ i.e. x & y are perfectly correlated.

② $b_{yx} = r\frac{\sigma_y}{\sigma_x} = 1 \cdot \frac{2\sqrt{2}}{\sqrt{2}} = 2$

③ $b_{xy} = r\frac{\sigma_x}{\sigma_y} = 1 \cdot \frac{\sqrt{2}}{2\sqrt{2}} = \frac{1}{2}$

④ R. line y on x :-

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 7 = 2(x - 3)$$

$$\boxed{y = 2x + 1}$$

⑤ R. line x on y :-

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 3 = \frac{1}{2}(y - 7)$$

$$\boxed{x = \frac{y}{2} - \frac{1}{2}}$$

Note: $(\bar{x}, \bar{y}) = (3, 7)$ satisfies both the lines

Q: If $\bar{x}=10$, $\bar{y}=90$, $\sigma_x=3$, $\sigma_y=12$ & $r=0.8$

then find R. Line y on x and x on y ?

Sol: $b_{yx}=r\frac{\sigma_y}{\sigma_x}=0.8\left(\frac{12}{3}\right)=3.2$

R line y on x :-

$$y-\bar{y}=b_{yx}(x-\bar{x})$$

$$y-90=3.2(x-10)$$

$$y=3.2x+58$$

Now $b_{xy}=r\frac{\sigma_x}{\sigma_y}=0.8\left(\frac{3}{12}\right)=0.2$

Now R. line x on y :-

$$x-\bar{x}=b_{xy}(y-\bar{y})$$

$$x-10=0.2(y-90)$$

$$x=0.2y-8$$

Note- $(\bar{x},\bar{y})=(10,90)$ lies on both lines

Note :- $x - 10 = 0.2(y - 90)$

$x - 10 = 0.2y - 0.2 \times 90$

$x = 0.2y - 45 + 10$??

$x = 0.2y - 35$



(*) $0.2 \times 90 = \frac{1}{5} \times 90 = 18$



Q If x & y are Independent then angle b/n two R. lines ?

(a) 0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{3\pi}{4}$

Sol: for Ind. R. Variables, $r = 0$

So $\tan\theta = \left(\frac{1-r^2}{r^2}\right)\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$

$\tan\theta = \left(\frac{1}{0}\right)\left(\downarrow\right) = \infty$

$\theta = \pi/2$

Q: Find π, b_{yx}, b_{xy}, R.Line y on x & R.Line x on y for :

HW Q

$x=$	1	2	3	4	5	6	7
$y=$	9	8	10	12	11	13	14

Ans: $b_{yx} = b_{xy} = r = 0.929$ (i.e Both b_{yx} & b_{xy} are less than 1)

& R line y on x is $y = 0.929x + 7.28$

R line x on y is $x = 0.929y - 6.219$

THANK - YOU